

Module 2: Intelligent Data Driven Applications on the Cloud

INTELLIGENT DATA DEVELOPMENT, JUNE 2026

Part 3: Cloud as a catalyst for data driven intelligence

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KEY CONCEPTS

- What is Cloud Computing? Cloud Timelines and Trends
- Cloud Computing
 - Compare and contrast cloud computing with traditional data centers
- Know the different cloud deployment models:
 - Public Cloud
 - Private Cloud
 - Hybrid Cloud
- Most Important Cloud Services
 - Storage
 - Databases
 - Compute
 - Machine Learning
 - Analytics

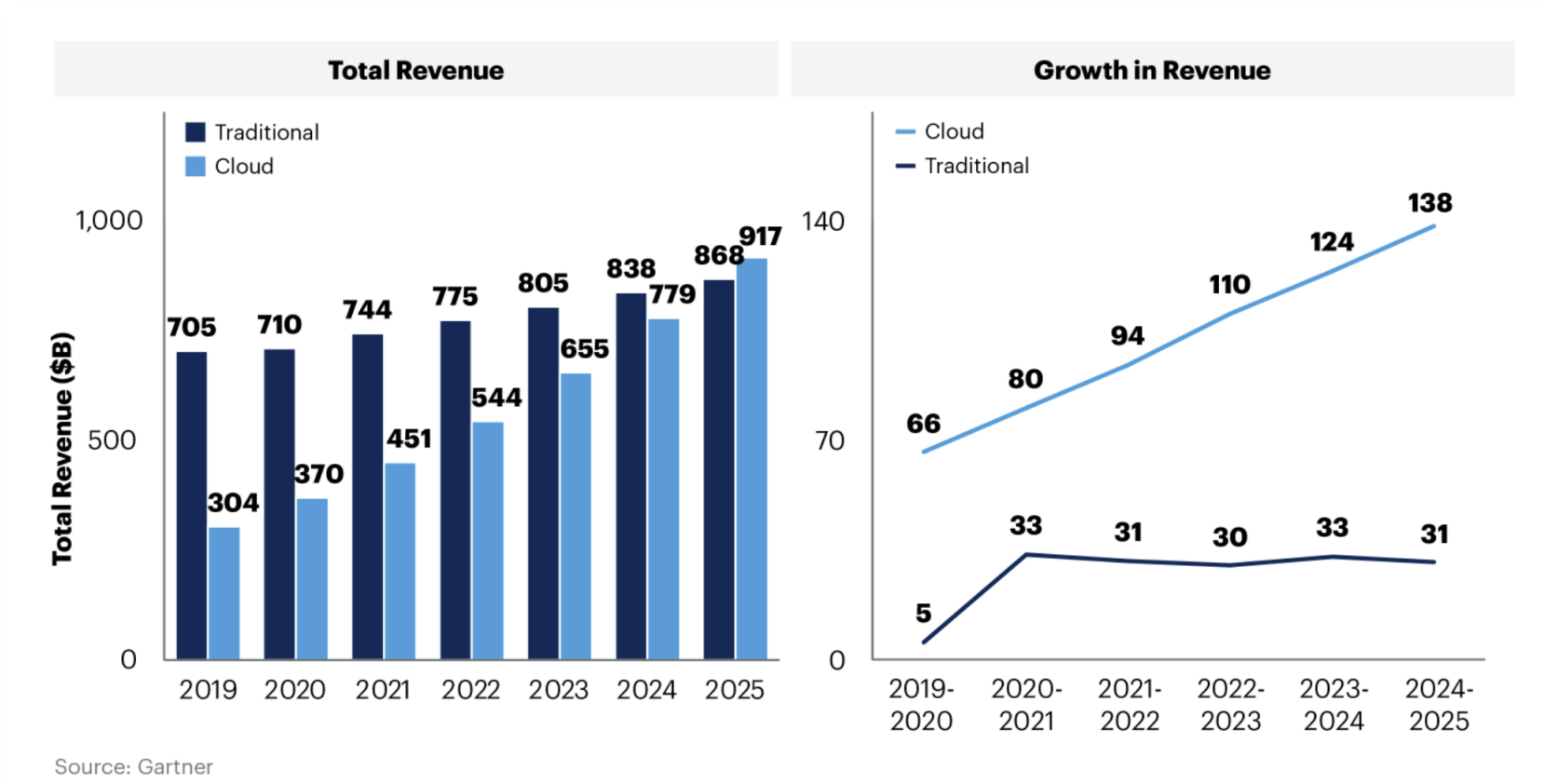
CLOUD COMPUTING

- **WHAT IS CLOUD COMPUTING?**

“Cloud computing is the on-demand delivery of compute power, database storage, applications and other IT resources through a cloud services platform via the Internet with pay-as-you-go pricing.”

AMAZON WEB SERVICES

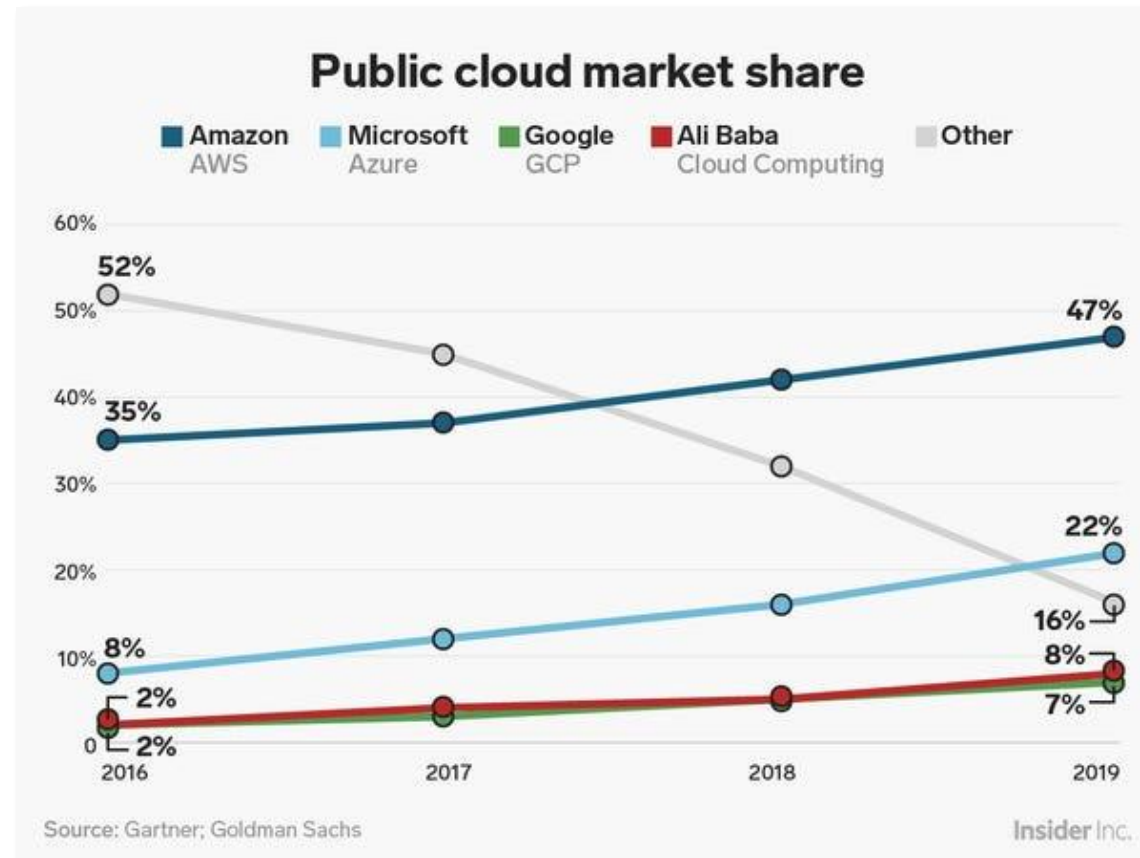
CLOUD TRENDS 1



“The shift to the cloud has only accelerated over the past years due to [COVID-19](#), as organizations responded to a new business and social dynamic. Technology and service providers that fail to adapt to the pace of cloud shift face increasing risk of becoming obsolete or, at best, being relegated to low-growth markets.”

[Michael Warrilow](#), Research VP at Gartner

CLOUD TRENDS 2



“The shift to the cloud has only accelerated over the past two years due to [COVID-19](#), as organizations responded to a new business and social dynamic. Technology and service providers that fail to adapt to the pace of cloud shift face increasing risk of becoming obsolete or, at best, being relegated to low-growth markets.”

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CLOUD TRENDS 3 – BEYOND THE BIG 4



WHY IS CLOUD SO POWERFUL

REAL WORLD SCENARIO

Consumer Good Company

- Launching a new demand forecasting process for the company.
- The demand planning process is focused on a geography and a Pilot is deployed in Spain.
- Looking to expand across all market units if successful in Spain.
- Securing funding for the initial infrastructure.

TRADITIONAL DATA CENTERS

Traditional Data Center - facility/space of networked computers and associated components (like telecommunications and storage) which helps business and organisations to function a large amount of data

Characteristics:

- Large up-front investment
- High maintenance cost (done by data center employees)
- Scaling up requires huge investments (forecasting demand is difficult)
- Deploying new data centers and servers takes time (slow provisioning)
- Owning security and compliance
- The organization's developers are trusted for the data stored in data centers
- Performance is less than compared to investment
- It is easily customizable without any hard plan
- It may and may not require an internet connection
- Data Centers require experienced developers to operate and are considered not a viable option



MareNostrum SuperComputing Center, Barcelona

CLOUD VS TRADITIONAL DATACENTERS

Characteristics:

- Less upfront investment
- Less maintenance cost since is the service providers maintain it
- Benefit from massive economies of scale - Scaling up required less amount of investment and deploying new data centers takes minutes (stop guessing capacity)
- The service provider owns the security of the cloud, you own security in the cloud
- Third-Party needs to be trusted for the organization's data to be stored
- Performance is huge as compared with investment
- It requires a plan to customize the cloud
- It requires a stable internet connection to provide the function
- Cloud is easy to operate and is considered a viable option



Amazon Data Center

WHY IS CLOUD SO POWERFUL

WHAT IMPACT DOES THE CLOUD HAVE ON ALL BUSINESSES?

- **Access to almost infinitely scalable computing power data storage, without the need for an expensive investment upfront.**
- Cloud is **redefining roles in the IT industry**, it's merging roles together.
- Businesses can now spend their **time focused on what differentiates them from their competitors and what provides value to their customers.**

CLOUD DEPLOYMENT MODELS



PRIVATE CLOUD

Cloud service provided by a third-party provider, hardware is only used by a single company, which many times owns the hardware and data center

- Capital cost
- Lack of elasticity
- Potential latency
- High overhead
- High Security
- Fully customizable

Top Vendors:

- HPE
- VMWare
- Dell
- IBM/ RedHat
- Palantir



PUBLIC CLOUD

Cloud service provided by a third-party provider, hardware can be shared amongst multiple clients

- No capital cost
- Elastic - infinite scalability
- No latency
- Low overhead
- Faster and simpler deployments

Top Vendors:

- AWS
- Microsoft Azure
- Google Cloud



HYBRID CLOUD

Cloud service provided as a combination between public and private cloud with automation and orchestration between the two

- Some capital cost
- Adds elasticity and flexibility
- Often used to transition to public cloud

Top Vendors:

- A combination of private and public cloud providers

AWS KEY SERVICES

- **Amazon Web Services** offers global cloud-based products that **enable organizations to move faster, lower IT costs, and scale**
- **Currently 238 products** split in **21 categories** - 11 products launched at re:Invent 2022



ANALYTICS



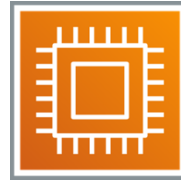
APPLICATION
INTEGRATION



BLOCKCHAIN



AWS COST
MANAGEMENT



COMPUTE



CONTAINERS



DATABASE



DEVELOPER
TOOLS



END USER
COMPUTER



FRONT END WEB &
MOBILE



IoT



MACHINE
LEARNING



MANAGEMENT &
GOVERNANCE



MEDIA
SERVICES



MIGRATION &
TRANSFER



NETWORK & CONTENT
DELIVERY



QUANTUM
TECHNOLOGY



ROBOTICS



SATELLITE



SECURITY, IDENTITY &
COMPLIANCE



STORAGE

AWS KEY SERVICES - FOCUS

- Amazon Web Services offers global cloud-based products that enable organizations to move faster, lower IT costs, and scale.
- Currently 226 products split in 21 categories - 7 products launched at re:Invent 2021



AWS KEY SERVICES - FOCUS



STORAGE

A service that allows **object storage** and offers industry-leading scalability, data availability, security, and performance.

Service Examples:

- Amazon EFS
- Amazon EBS
- AWS Snowball
- AWS Snowmobile
- **Amazon S3**

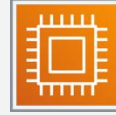


DATABASE

A service that enables engines to support diverse data models, including relational, key-value, document, in-memory, graph, time series, wide column, and ledger **databases**.

Service Examples:

- Amazon Aurora
- **Amazon DynamoDB**
- Amazon Neptune
- **Amazon RDS**



COMPUTE

A service that enables to leverage **cloud-based virtual machines** for workloads. This could be serving web content to visitors, running a database, or calculating statistics from a data set.

Service Examples:

- **Amazon EC2**
- **AWS Lambda**
- AWS Outposts (on prem private cloud)



MACHINE LEARNING

A service that empowers ML teams to prepare and process data, streamlines cross-team collaboration, and **standardizes the full ML lifecycle** from experimentation to production.

Service Examples:

- Amazon Forecast
- **Amazon SageMaker**
- Amazon Transcribe
- Amazon Translate
- AWS DeepComposer



ANALYTICS

A service that enables data science teams to **deploy end-to-end analytics solutions** from data lake formation, to data storage and transport, data warehouse and visualization.

Service Examples:

- AWS Lake Formation
- AWS Glue
- Amazon Kinesis
- **Amazon Redshift**
- Amazon QuickSight

CLOUD STORAGE – AWS S3



Amazon Simple Storage
Service (Amazon S3)

- S3 stands for **Simple Storage Service (S3)**
- One of the oldest AWS services - launched in **2006** – and one of the most feature-rich storage platform available
- Allows the storage and retrieval of **any amount of data** (up to 5T)
- It is easy to set up and **easy-to-use**
- **Fast and cheap**: quickly store and retrieve any amount of data from anywhere on the web at an economical storage cost
- It is **secure** - allows encryption
- Provides **different storage classes for different use cases**

CLOUD STORAGE - S3



Amazon Simple Storage
Service (Amazon S3)

- It is **object based**. Think of objects just as files (i.e. Word documents, movies, pictures, CSVs, parquet files).
- Objects consist of the following:
 - **Key** – name of the object
 - **Value** – data
 - **Version**
 - **Metadata** – data about data
 - **Sub-resources** (access control lists, bucket policies)
- Files are stored in **buckets**. A bucket is a folder where you put your files.

CLOUD COMPUTE– AWS LAMBDA



AWS Lambda

- **Serverless architecture** is an approach to software design that allows developers to build and run services without having to manage the underlying infrastructure.

Fundamental serverless terms:

- **Invocation** - Single function execution
- **Duration** - Time taken for a lambda function to execute
- **Cold Start** - Latency when a function is triggered for the first time/ after an inactivity period
- **Concurrency Limit** - The number of function instances that can run simultaneously in one region
- **Timeout** - The amount of time that a cloud provider allows a function to run before terminating it – 15mins

Use Cases:

- Trigger-based tasks
- Building RESTful APIs
- Asynchronous processing
- Security checks
- Continuous Integration (CI) and Continuous Delivery (CD)

HANDS ON SESSION

INTELLIGENT DATA DEVELOPMENT, JUNE 2026

HANDS-ON CLOUD LAB: DEPLOY A FORECASTING MODEL ON AWS – 90MINS

Session Goal: Learn to **deploy a forecasting model** using step by step instructions and **Infrastructure as Code (IaC)** on AWS. You'll build, run, and test a simple ML workflow — all in the cloud.

What We'll Do

We'll work together to set up a real forecasting pipeline in the cloud:

1. You'll create your own AWS user, deploy cloud storage (S3)
2. Run a forecasting model using Lambda (a cloud function)
3. Use Infrastructure as Code (CloudFormation) to launch everything with just one command — giving you hands-on experience with how real-world data science projects are deployed."

What You'll Walk Away With

- You'll **train a model** in the cloud and **see your forecast results**
- You'll **upload and download files from AWS** (just like a data scientist)
- You'll understand how cloud tools **work together** in real projects
- You'll learn how to **deploy a project with code, not clicks** (just like pros do)

HANDS-ON CLOUD LAB: STEPS

1. Connect to an AWS account using your IAM user – 10 mins
2. Create an S3 bucket like idd-class-folder and load data.csv – 10 mins
3. Create a Lambda function by hand – 20 mins
4. Link the Lambda to S3 and give it proper permissions – 15 mins
5. Load data from s3 and save data in csv format into s3 – 20 mins
6. Launch all above with a CloudFormation template – 15 mins

HANDS ON SESSION DETAILS

AWS account: <https://idd-class.signin.aws.amazon.com/console>

Access Credentials:

- **Users:** 30 generic users created "idd2026student1" to "idd2026student30"
- **Password:** Initial Password "**ChangeMeNow_2024!**"

S3 Bucket with the following structure:

```
idd-class-folder/  
├─ cloud-formation-templates/  ← Shared: read-only for all students  
├─ data-input/                 ← Shared: read-only for all students  
└─ data-outputs/              ← Personal subfolders (studentN only)  
    ├─ student1/  
    ├─ student2/  
    ├─ ...  
    └─ student30/
```

- Each student can access only their results.
- Shared resources are readable by all.
- No one can access or overwrite another student's output.

HANDS ON – STEP 1 - CONNECT TO AN AWS ACCOUNT USING AN IAM USER

Step-by-step:

1. Go to the login link: **<https://idd-class.signin.aws.amazon.com/console>**
2. Log in using the credentials:
 - **Username:** studentuser1 (or your assigned name)
 - **Password:** the one you received
3. On first login, set a **new password** you'll remember. A password reset screen will appear. You will need to add the old password provided and add your new one

You're now inside your AWS account!

IAM user sign in ⓘ

Account ID or alias [\(Don't have?\)](#)

idd-class

☐ Remember this account

IAM username

idd2026student1|

Password

.....

☐ Show Password [Having trouble?](#)

Sign in

Sign in using root user email

Create a new AWS account

Password reset ⓘ

Your account (223947838) expired or requires a reset

To continue, please verify new password for student

Old Password

••••••••

☐ Show Password

New Password

••••••••••

☐ Show Password

Confirm New Password

••••••••••

☐ Show Password

Matches

Confirm Password Change

[Sign in to a different account](#)

Password resets are managed by your administrator. In most cases your password should contain:

- Minimum password length is 8 characters
- Include a minimum of three of the following mix of character types: uppercase, lowercase, numbers, and ! @ # \$ % ^ & * () _ + - = [] { } ' ' '
- Must not be identical to your AWS account name or email address

HANDS ON – STEP 2 - CREATE AN S3 BUCKET

We'll now create a cloud folder (S3 bucket) with versioning, encryption, and private access.

Steps:

1. Search for **S3** in the top search bar, click **S3**
2. Click **Create bucket**
3. Name your bucket:
 - **student1**-folder (replace with your ID), your bucket needs to start with the student1 prefix
4. Choose your region (e.g., us-east-1)
5. **Under settings:**
 - **Block Public Access:** Keep it ON (all boxes checked)
 - **Bucket Versioning:** Enable
 - **Default Encryption:** Enable (use **SSE-S3**)
6. Click **Create bucket**

HANDS ON – STEP 3 - CREATE A LAMBDA FUNCTION BY HAND

Step 1: Create an IAM Role for Lambda

1. Go to **IAM > Roles**
2. Click **Create role**
3. Choose **AWS service** → **Lambda**
4. Attach the following policies:
 1. AmazonS3FullAccess (or scoped-down access)
 2. AWSLambdaBasicExecutionRole
5. Name the role (e.g. LambdaS3Role) and create it.

Step 2: Create the Lambda Function

1. Go to **Lambda** → **Create function**
2. Choose:
 1. Name: ForecastingFunction
 2. Runtime: **Python 3.X**
 3. Execution role: Use existing role → select LambdaS3PandasRole
3. Create the function

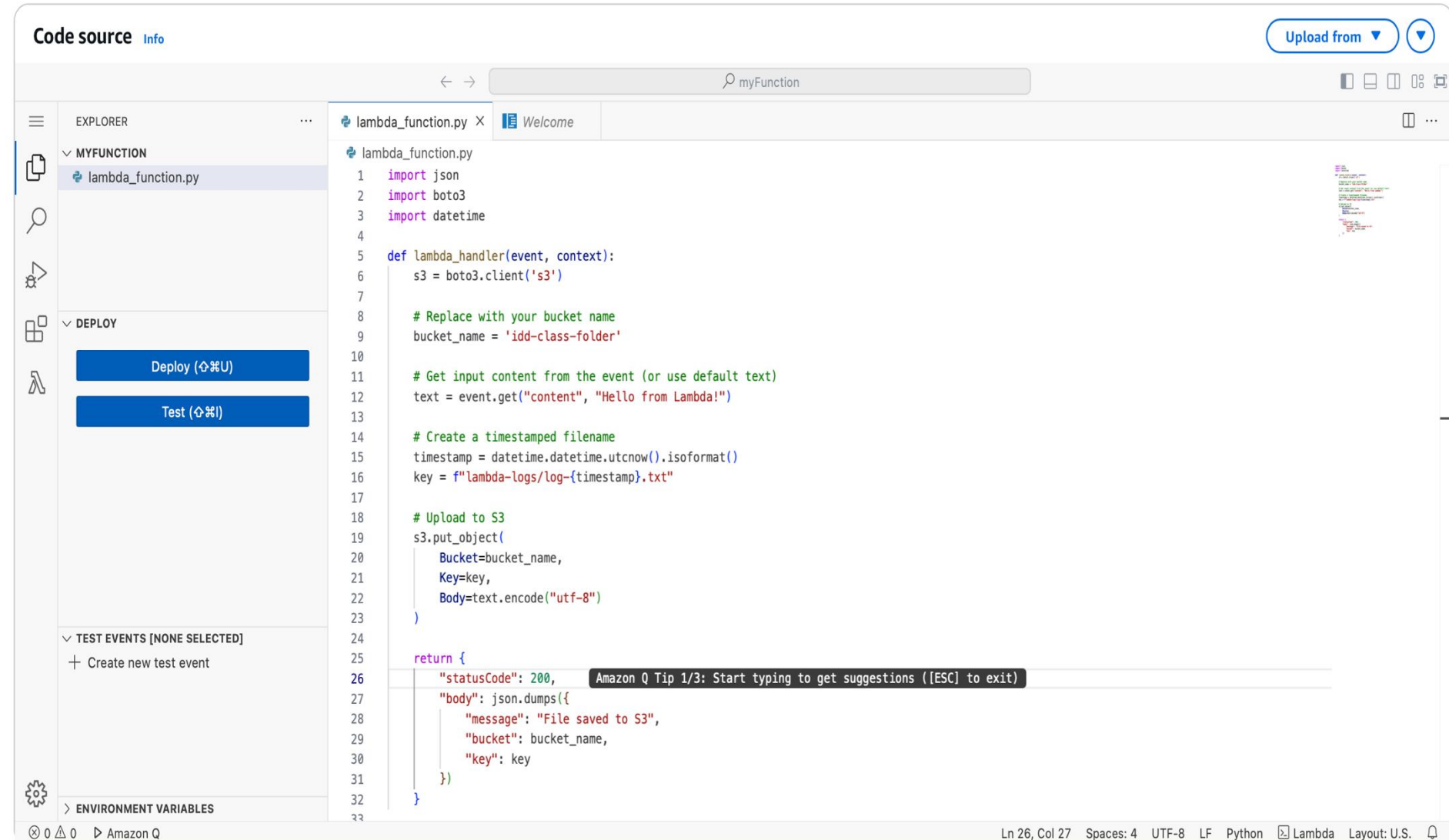
HANDS ON – STEP 3 - CREATE A LAMBDA FUNCTION BY HAND

Step 3: Edit Lambda Code

Click **Code** and paste the following:
Replace your-input-bucket-name and file keys with your actual S3 path.

Step 4: Test the Lambda

1. Click **Test** → Create a test event (can be empty if hardcoded paths are used)
2. Run the Lambda
3. Check your S3 bucket to confirm the new file is saved



The screenshot shows the AWS Lambda console's 'Code source' page for a function named 'myFunction'. The interface includes a left-hand sidebar with 'EXPLORER' and 'DEPLOY' sections. The 'EXPLORER' section shows a file named 'lambda_function.py' under the 'MYFUNCTION' folder. The 'DEPLOY' section has buttons for 'Deploy (⌘%U)' and 'Test (⌘%I)'. The main area displays the Python code for 'lambda_function.py'. The code imports 'json', 'boto3', and 'datetime', and defines a 'lambda_handler' function. The function uses 'boto3' to upload a file to an S3 bucket. Comments in the code indicate where to replace the bucket name and the file key. The status bar at the bottom shows 'Ln 26, Col 27', 'Spaces: 4', 'UTF-8', 'LF', 'Python', 'Lambda', and 'Layout: U.S.'.

```
1 import json
2 import boto3
3 import datetime
4
5 def lambda_handler(event, context):
6     s3 = boto3.client('s3')
7
8     # Replace with your bucket name
9     bucket_name = 'add-class-folder'
10
11     # Get input content from the event (or use default text)
12     text = event.get("content", "Hello from Lambda!")
13
14     # Create a timestamped filename
15     timestamp = datetime.datetime.utcnow().isoformat()
16     key = f"lambda-logs/log-{timestamp}.txt"
17
18     # Upload to S3
19     s3.put_object(
20         Bucket=bucket_name,
21         Key=key,
22         Body=text.encode("utf-8")
23     )
24
25     return {
26         "statusCode": 200,
27         "body": json.dumps({
28             "message": "File saved to S3",
29             "bucket": bucket_name,
30             "key": key
31         })
32     }
```


HANDS ON – STEP 3 - CREATE A LAMBDA FUNCTION BY HAND

Step 5: Package pandas for Lambda - optional

Lambda has a size limit for inline editing, and pandas isn't included by default. You must **deploy with a layer** or **zip upload**.

Option A: Use a Prebuilt Lambda Layer (Recommended for Console Users)

Use a public pandas layer:

1. Visit [AWS Lambda Layer Library](#) → Find the **pandas layer ARN** for your region (e.g. arn:aws:lambda:us-east-1:770693421928:layer:Klayers-p39-pandas:1)
2. Save the ARN for later

Step 6: Add pandas Layer to Lambda - optional

1. Open the Lambda function → Click **Layers** → **Add a layer**
2. Choose **Specify an ARN**
3. Paste your pandas layer ARN and click **Add**

LAMBA CODE

```
import json
import boto3
import datetime
import pandas as pd
from io import StringIO

def lambda_handler(event, context):
    s3 = boto3.client('s3')

    # Replace with your actual bucket name
    bucket_name = 'student1-bucket-idd'
    input_key = 'input.csv'

    try:
        # Read CSV from S3
        response = s3.get_object(Bucket=bucket_name, Key=input_key)
        content = response['Body'].read().decode('utf-8')
        df = pd.read_csv(StringIO(content))

        # Do any transformation here (this is just a placeholder)
        df_processed = df.copy()

        # Prepare to write back to S3
        timestamp = datetime.datetime.utcnow().isoformat()
        output_key = f"processed/output_{timestamp}.csv"
        csv_buffer = StringIO()
        df_processed.to_csv(csv_buffer, index=False)

        # Upload to S3
        s3.put_object(
            Bucket=bucket_name,
            Key=output_key,
            Body=csv_buffer.getvalue().encode('utf-8')
        )
```

```
        return {
            "statusCode": 200,
            "body": json.dumps({
                "message": "CSV processed and saved to S3",
                "bucket": bucket_name,
                "output_key": output_key
            })
        }

    except Exception as e:
        return {
            "statusCode": 500,
            "body": json.dumps({
                "error": str(e)
            })
        }
```